

Topic 6 - Testlet 10

Question #1

Topic 6

Introductory Info

Case Study -

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All Information tab, note that the information displayed is identical to the information displayed on the subsequent tabs. When you are ready to answer a question, click the Question button to return to the question.

General Overview -

Northwind Traders is a specialty food import company.

The company recently implemented Power BI to better understand its top customers, products, and suppliers.

Business Issues -

The sales department relies on the IT department to generate reports in Microsoft SQL Server Reporting Services (SSRS). The IT department takes too long to generate the reports and often misunderstands the report requirements.

Existing Environment. Data Sources

Northwind Traders uses the data sources shown in the following table.

Name	Type	Data size
Source1	Azure SQL database	2 GB
Source2	Microsoft Excel spreadsheet	5 MB

Source2 is exported daily from a third-party system and stored in Microsoft SharePoint Online.

Existing Environment. Customer Worksheet

Source2 contains a single worksheet named Customer Details. The first 11 rows of the worksheet are shown in the following table.

CustomerID	CustomerCRMID	CompanyName	Address	City	Region	PostalCode	Country	Phone
1	ALFKI	Alfreds Futterkiste	Obere Str. 57	Berlin	DE	12209	Germany	030-0074321
2	ANATR	Ana Trujillo Emparedados y helados	Avda. de la Constitución 2222	México D.F.	MX	5021	Mexico	(5) 555-4729
3	ANTON	Antonio Moreno Taquería	Mataderos 2312	México D.F.	MX	5023	Mexico	(5) 555-3932
4	AROUT	Around the Horn	120 Hanover Sq.	London	UK	WA1 1DP	UK	(171) 555-7788
5	BERGS	Berglunds snabbköp	Berguvsvägen 8	Luleå	SWE	S-958 22	Sweden	0921-12 34 65
6	BLAUS	Blauer See Delikatessen	Forsterstr. 57	Mannheim	DE	68306	Germany	0621-08460
7	BLONP	Blondesddsl père et fils	24, place Kléber	Strasbourg	FRA	67000	France	88.60.15.31
8	BOLID	Bólido Comidas preparadas	C/ Araquil, 67	Madrid	SPN	28023	Spain	(91) 555 22 82
9	BONAP	Bon app'	12, rue des Bouchers	Marseille	FRA	13008	France	91.24.45.40
10	BOTTM	Bottom-Dollar Markets	23 Tsawassen Blvd.	Tsawassen	BC	T2F 8M4	Canada	(604) 555-4729

All the fields in Source2 are mandatory.

The Address column in Customer Details is the billing address, which can differ from the shipping address.

Existing Environment. Azure SQL Database

Source1 contains the following tables:

Orders

Products

Suppliers

Categories

Order Details

Sales Employees

The Orders table contains the following columns.

Name	Is nullable	Data type	Example value	Key
OrderID	No	Int	10248	Primary key
CustomerID	Yes	NCHAR	VINET	Not applicable
OrderDate	Yes	Date	2021-01-04	Not applicable
RequiredDate	Yes	Date	2021-02-01	Not applicable
ShippedDate	Yes	Date	2021-01-16	Not applicable
Freight	Yes	Decimal	32.38	Not applicable
ShipName	Yes	NVARCHAR	Vins et alcools Chevalier	Not applicable
ShipAddress	Yes	NVARCHAR	59 rue de l'Abbaye	Not applicable
ShipCity	Yes	NVARCHAR	Reims	Not applicable
ShipRegion	Yes	NVARCHAR	FRA	Not applicable
ShipPostalCode	Yes	NVARCHAR	51100	Not applicable
ShipCountry	Yes	NVARCHAR	France	Not applicable

The Order Details table contains the following columns.

Name	Is nullable	Data type	Example value	Key
OrderID	No	Int	10248	Foreign key to Orders
ProductID	No	Int	11	Foreign key to Products
UnitPrice	No	Decimal	14	Not applicable
Quantity	No	Smallint	12	Not applicable
Discount	No	Decimal	0.15	Not applicable

The address in the Orders table is the shipping address, which can differ from the billing address.

The Products table contains the following columns.

Name	Is nullable	Data type	Example value	Key
ProductID	No	Int	11	Primary key
ProductName	No	NVARCHAR	Queso Cabrales	Not applicable
SupplierID	Yes	Int	5	Foreign key to Suppliers
CategoryID	Yes	Int	4	Foreign key to Categories
QuantityPerUnit	Yes	NVARCHAR	1 kg pkg.	Not applicable
Discontinued	No	Bit	0	Not applicable

The Categories table contains the following columns.

Name	Is nullable	Data type	Example value	Key
CategoryID	No	int	4	Primary key
CategoryName	No	nvarchar	Dairy Products	Not applicable
Description	Yes	nvarchar	Cheeses	Not applicable

The Suppliers table contains the following columns.

Name	Is nullable	Data type	Example value	Key
SupplierID	No	Int	5	Primary key
CompanyName	No	NVARCHAR	Cooperativa de Quesos 'Las Cabras'	Not applicable
Address	Yes	NVARCHAR	Calle del Rosal 4	Not applicable
City	Yes	NVARCHAR	Oviedo	Not applicable
Region	Yes	NVARCHAR	Asturias	Not applicable
PostalCode	Yes	NVARCHAR	33007	Not applicable
Country	Yes	NVARCHAR	Spain	Not applicable
Phone	Yes	NVARCHAR	(98) 598 76 54	Not applicable

The Sales Employees table contains the following columns.

Name	Is nullable	Data type	Example value	Key
EmployeeID	No	Int	1	Primary key
LastName	No	NVARCHAR	Davolio	Not applicable
FirstName	No	NVARCHAR	Nancy	Not applicable
Title	Yes	NVARCHAR	Sales Representative	Not applicable
HireDate	Yes	Date	2015-02-01	Not applicable
Region	Yes	NVARCHAR	WA	Not applicable
Country	Yes	NVARCHAR	USA	Not applicable
EmailAddress	No	NVARCHAR	ndavolio@northwindtraders.com	Not applicable

Each employee in the Sales Employees table is assigned to one sales region. Multiple employees can be assigned to each region.

Requirements. Report Requirements

Northwind Traders requires the following reports:

Top Products

Top Customers

On-Time Shipping

The Top Customers report will show the top 20 customers based on the highest sales amounts in a selected order month or quarter, product category, and sales region.

The Top Products report will show the top 20 products based on the highest sales amounts sold in a selected order month or quarter, sales region, and product category. The report must also show which suppliers provide the top products.

The On-Time Shipping report will show the following metrics for a selected shipping month or quarter:

The percentage of orders that were shipped late by country and shipping region

Customers that had multiple late shipments during the last quarter

Northwind Traders defines late orders as those shipped after the required shipping date.

The warehouse shipping department must be notified if the percentage of late orders within the current month exceeds 5%.

The reports must show historical data for the current calendar year and the last three calendar years.

Requirements. Technical Requirements

Northwind Traders identifies the following technical requirements:

A single dataset must support all three reports.

The reports must be stored in a single Power BI workspace.

Report data must be current as of 7 AM Pacific Time each day.

The reports must provide fast response times when users interact with a visualization.

The data model must minimize the size of the dataset as much as possible, while meeting the report requirements and the technical requirements.

Requirements. Security Requirements

Access to the reports must be granted to Azure Active Directory (Azure AD) security groups only. An Azure AD security group exists for each department.

The sales department must be able to perform the following tasks in Power BI:

Create, edit, and delete content in the reports.

Manage permissions for workspaces, datasets, and reports.

Publish, unpublish, update, and change the permissions for an app.

Assign Azure AD groups role-based access to the reports workspace.

Users in the sales department must be able to access only the data of the sales region to which they are assigned in the Sales Employees table.

Power BI has the following row-level security (RLS) Table filter DAX expression for the Sales Employees table.

[EmailAddress] = USERNAME()

RLS will be applied only to the sales department users. Users in all other departments must be able to view all the data.

Question

You need to create the dataset.

Which dataset mode should you use?

- A. Import
- B. DirectQuery
- C. Composite
- D. live connection

Correct Answer: C

The three dataset modes are:

Import -

DirectQuery -

Composite -

Composite mode -

Composite mode can mix Import and DirectQuery modes, or integrate multiple DirectQuery data sources. Models developed in Composite mode support configuring the storage mode for each model table. This mode also supports calculated tables (defined with DAX).

The table storage mode can be configured as Import, DirectQuery, or Dual. A table configured as Dual storage mode is both Import and DirectQuery, and this setting allows the Power BI service to determine the most efficient mode to use on a query-by-query basis.

Note: A single dataset must support all three reports.

The data model must minimize the size of the dataset as much as possible, while meeting the report requirements and the technical requirements.

Northwind Traders requires the following reports:

Top Products -

Top Customers -

On-Time Shipping -

Incorrect:

* DirectQuery mode is an alternative to Import mode. Models developed in DirectQuery mode don't import data. Instead, they consist only of metadata defining the model structure. When the model is queried, native queries are used to retrieve data from the underlying data source.

* Import

However, while there are compelling advantages associated with Import models, there are disadvantages, too:

The entire model must be loaded to memory before Power BI can query the model, which can place pressure on available capacity resources, especially as the number and size of Import models grow

Community vote distribution

C (53%)

A (47%)

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Question

You need to configure access for the sales department users. The solution must the meet the security requirements.

What should you do?

- A. Share each report to the Azure Active Directory group of the sales department.
- B. Add the Azure Active Directory group of the sales department as an Admin of the reports workspace.
- C. Distribute an app to the users in the Azure Active Directory group of the sales department.
- D. Add the sales department as a member of the reports workspace.

Correct Answer: *B*

Access to the reports must be granted to Azure Active Directory (Azure AD) security groups only. An Azure AD security group exists for each department.

The sales department must be able to perform the following tasks in Power BI:

- ☒ Create, edit, and delete content in the reports.
- ☒ Manage permissions for workspaces, datasets, and reports.
- ☒ Publish, unpublish, update, and change the permissions for an app.
- ☒ Assign Azure AD groups role-based access to the reports workspace.

Community vote distribution

C (55%) D (34%) 5%

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Existing Environment. Azure SQL Database

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- Orders
- Products
- Suppliers
- Categories
- Order Details
- Sales Employees

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Question

HOTSPOT -

You need to create a solution to meet the notification requirements of the warehouse shipping department.
What should you do? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Hot Area:

Answer Area

Populate a

report

bookmark

dashboard

by using a card visualization that shows the percentage of late

orders in the current month, and then configure a

data alert.

phone view.

subscription.

Correct Answer:

Answer Area

Populate a

report

bookmark

dashboard

by using a card visualization that shows the percentage of late

orders in the current month, and then configure a

data alert.

phone view.

subscription.

Box 1: dashboard -
The warehouse shipping department must be notified if the percentage of late orders within the current month exceeds 5%.
You can set alerts to notify you when data in your dashboards changes beyond limits you set.

Box 2: data alert -
Reference:
<https://docs.microsoft.com/en-us/power-bi/create-reports/service-set-data-alerts>

Topic 7 - Testlet 2

Question #1

Topic 7

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Overview -

Contoso, Ltd. is a manufacturing company that produces sports equipment. Contoso holds quarterly board meetings for which financial analysts manually prepare

Microsoft Excel reports, including balance sheets and profit and loss statements for each of the company's four business units.

Existing Environment -

Data and Sources -

Data for the reports comes from the sources shown in the following table.

Data type	Description
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Microsoft Dynamics 365 Business Central	Summary balance sheet data and product catalog data

The balance sheet data is unrelated to the profit and loss results other than they both relate to dates.

Balance Sheet Data -

The balance sheet data is imported and includes the final monthly balances of each account in the format shown in the following table.

AccountCategory	Account	Month	Year	BalanceAmount
Current assets	Cash and cash equivalents	3	2020	20,289
Current assets	Inventories	3	2020	4,855
Long-term liabilities	Long-term debt	3	2020	50,207
Current assets	Cash and cash equivalents	2	2020	28,209
Current assets	Inventories	2	2020	5,845
Long-term liabilities	Long-term debt	2	2020	49,887
Current assets	Cash and cash equivalents	1	2020	25,567
Current assets	Inventories	1	2020	65,998
Long-term liabilities	Long-term debt	1	2020	46,124

The balance sheet data always includes a row for each account for each month.

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The product catalog shows how products roll up to product categories, which roll up to the business units. The product list is provided in the

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Revenue data is provided at the date and product level. Expense data is provided at the date and department level.

Business Issues -

Historically, it has taken two analysts a week to prepare the reports for the quarterly board meetings. Also, there is usually at least one issue each quarter where a value in a report is wrong because of a bad cell reference in an Excel formula. On occasion, there are conflicting results in the reports because the products and departments that roll up to each business unit are not defined consistently.

Requirements -

Planned Changes -

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▪

The reports must be updated with the latest data by 5 AM each day.

Question

You need to create the relationship between the product list and the revenue results. The solution must minimize the time it takes to render visuals.

What should you set as the relationship cardinality?

- A. One to one
- B. Many to many
- C. Many to one
- D. One to many

Correct Answer: D

One product in the product list can occur many times in the revenue results.

Note 1: One to many (1:*) : In a one-to-many relationship, the column in one table has only one instance of a particular value, and the other related table can have more than one instance of a value.

Note 2:

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Reference:

<https://docs.microsoft.com/en-us/power-bi/transform-model/desktop-create-and-manage-relationships>

Community vote distribution

D (92%)

8%

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Question

HOTSPOT -

You need to create a measure that returns the percent of revenue by product category.

How should you complete the measure? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Hot Area:

Category Revenue Contribution =
VAR AllCategoryRev =

▼

(SUM ([Revenue]),

ALL

ALLEXCEPT

CALCULATE

DIVIDE

FILTER

▼

(ProductList [ProductCategory]))

ALL

ALLEXCEPT

CALCULATE

DIVIDE

FILTER

RETURN

▼

(SUM ([Revenue]), AllCategoryRev)

ALL

ALLEXCEPT

CALCULATE

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FILTER

Category Revenue Contribution =
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▼

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ALL

ALLEXCEPT

CALCULATE

DIVIDE

FILTER

RETURN

▼

(SUM([Revenue]), AllCategoryRev)

ALL

ALLEXCEPT

CALCULATE

DIVIDE

FILTER

Correct Answer:

Box 1: CALCULATE -
CALCULATE evaluates an expression in a modified filter context.
Syntax: CALCULATE(<expression>[, <filter1> [, <filter2> [, ...]])

Box 2: REMOVEFILTERS -
REMOVEFILTERS clear filters from the specified tables or columns.

Box 3: DIVIDE -
DIVIDE performs a division.
Example: MEASURE FactInternetSales[%Sales] = DIVIDE([TotalSales], CALCULATE([TotalSales],REMOVEFILTERS()))
Note: The RETURN keyword consumes variables defined in previous VAR statements.
Reference:
<https://docs.microsoft.com/en-us/dax/calculate-function-dax>
<https://docs.microsoft.com/en-us/dax/removefilters-function-dax> <https://dax.guide/st/return/>

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Question

DRAG DROP -

Once the profit and loss dataset is created, which four actions should you perform in sequence to ensure that the business unit analysts see the appropriate profit and loss data? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.

Select and Place:

Actions

- From powerbi.com, assign the analysts the Contributor role to the workspace.
- From Power BI Desktop, add a Table Filter DAX Expression to the roles.
- From powerbi.com, add role members to the roles.
- From Power BI Desktop, create four roles.
- From Power BI Desktop, publish the dataset to powerbi.com.

Answer Area

Correct Answer:

Actions

- From powerbi.com, assign the analysts the Contributor role to the workspace.
-
-
-
-

Answer Area

- From Power BI Desktop, create four roles.
- From Power BI Desktop, add a Table Filter DAX Expression to the roles.
- From powerbi.com, add role members to the roles.
- From Power BI Desktop, publish the dataset to powerbi.com.

Step 1: From BI Desktop, create four roles

You can define roles and rules within Power BI Desktop.

Step 2: From BI Desktop, add a Table Filter DAX Expression to the roles.

To define security roles, follow these steps.

1. From the Modeling tab, select Manage Roles.
2. Select Manage Roles
3. From the Manage roles window, select Create.
4. Select Create
5. Under Roles, provide a name for the role.
6. Under Tables, select the table to which you want to apply a DAX rule.
7. In the Table filter DAX expression box, enter the DAX expressions. This expression returns a value of true or false. For example: [Entity ID] = $\lambda \epsilon \text{Value} \lambda \epsilon$.

After you've created the DAX expression, select the checkmark above the expression box to validate the expression.

8. Select Save.

Step 3: From powerbi.com, add role members to the roles.

You can't assign users to a role within Power BI Desktop. You assign them in the Power BI service.

Step 4: From BI Desktop, publish the dataset to powerbi.com

Now that you're done validating the roles in Power BI Desktop, go ahead and publish your report to the Power BI service.

Reference:

<https://docs.microsoft.com/en-us/power-bi/enterprise/service-admin-rls>

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Question

Which DAX expression should you use to get the ending balances in the balance sheet reports?

- A. CALCULATE (SUM(BalanceSheet[BalanceAmount]), LASTDATE ('Date'[Date]))
- B. CALCULATE (SUM(BalanceSheet[BalanceAmount]), DATESQTD('Date'[Date]))
- C. FIRSTNONBLANK ('Date'[Date] SUM(BalanceSheet[BalanceAmount]))
- D. CALCULATE (MAX(BalanceSheet[BalanceAmount]), LASTDATE ('Date'[Date]))

Correct Answer: *B*

Use the CALCULATE, the SUM, and the DATESQTD functions.

DATESQTD returns a table that contains a column of the dates for the quarter to date, in the current context.

Syntax: DATESQTD(<dates>)

Note: At least one of the balance sheet reports in the quarterly reporting package must show the ending balances for the quarter, as well as for the previous quarter.

Reference:

<https://docs.microsoft.com/en-us/dax/datesqtd-function-dax>

Community vote distribution

A (62%)

B (38%)

Topic 8 - Testlet 3

Question #1

Topic 8

Introductory Info

Case Study -

This is a case study. Case studies are not timed separately. You can use as much exam time as you would like to complete each case. However, there may be additional case studies and sections on this exam. You must manage your time to ensure that you are able to complete all questions included on this exam in the time provided.

To answer the questions included in a case study, you will need to reference information that is provided in the case study. Case studies might contain exhibits and other resources that provide more information about the scenario that is described in the case study. Each question is independent of the other questions in this case study.

At the end of this case study, a review screen will appear. This screen allows you to review your answers and to make changes before you move to the next section of the exam. After you begin a new section, you cannot return to this section.

To start the case study -

To display the first question in this case study, click the Next button. Use the buttons in the left pane to explore the content of the case study before you answer the questions. Clicking these buttons displays information such as business requirements, existing environment and problem statements. If the case study has an

All Information tab, note that the information displayed is identical to the information displayed on the subsequent tabs. When you are ready to answer a question, click the Question button to return to the question.

Overview -

Litware, Inc. is an online retailer that uses Power BI.

Litware plans to leverage data from an Azure SQL database that stores data for the company's live e-commerce website.

Litware uses Azure Active Directory (Azure AD) to authenticate users.

Existing Environment. Sales Data

Litware has online sales data that has the SQL schema shown in the following table.

Table name	Column name	Data type
Sales_Region	region_id	Integer
	name	Varchar
Region_Manager	region_id	Integer
	manager_id	Integer
Sales_Manager	sales_manager_id	Integer
	name	Varchar
	username	Varchar
Manager	manager_id	Integer
	name	Varchar
Sales	sales_id	Integer
	sales_date_id	Integer
	sales_amount	Float
	customer_id	Integer
	sales_ship_date_id	Integer
	region_id	Varchar
Date	date_id	Integer
	date	Date
	month	Integer
	week	Integer
	year	Integer
Weekly>Returns	week_id	Integer
	total_returns	Float
	sales_region_id	Varchar
Targets	target_id	Integer
	sales_target	Decimal
	date_id	Integer
	region_id	Integer

In the Date table, the date_id column has a format of yyyyymmdd and the month column has a format of yyyyymm.